Southeast Asia Tables — Profitability Analysis

Global Superstore | Microsoft Excel Portfolio Project

Business Problem

The Southeast Asia Tables category generated $52,452 in sales but a loss of $18,618.31, resulting in a -35.50% profit margin.

Management needs to understand why the category is unprofitable, particularly whether the level of discounting is contributing to the losses, and which products are most affected.

The analysis therefore investigates the relationship between discount levels, sales, and profitability to identify opportunities to reduce losses and improve margins.

Project Objective

The objective of this analysis is to evaluate the impact of discounting on the profitability of Tables sold in Southeast Asia, identify the discount levels and products contributing most to losses, and provide actionable recommendations to improve profitability.

Dataset & Scope

Dataset: Global Superstore

Analysis Tool: Microsoft Excel

- Geographic Scope: Southeast Asia
- Product Category: Tables
- Primary Focus: Sales, profit, discount levels, and product-level profitability

The analysis uses the Global Superstore dataset to evaluate the profitability of the Tables category within the Southeast Asia region. The analysis focuses on how different discount levels affect profitability and identifies the products contributing most significantly to losses.

Data Preparation & Quality Checks

Before conducting the profitability analysis, the dataset was profiled and checked for data-quality issues.

Key validation checks included:

- 51,290 total transaction records were identified.
- 25,035 unique Order IDs and 4,873 unique customers were identified.
- 10,292 unique products were identified.
- No blank values were found in the key analytical fields: Sales, Profit, Quantity, Shipping Cost, Discount, Customer, Order ID, or Product ID.
- 12,544 transactions recorded negative profit, while 38,078 recorded positive profit and 668 recorded zero profit.
- Discount values ranged from 0% to 85%, while quantity ranged from 1 to 14 units.
- Shipping costs and sales values were also checked for unusual ranges and zero values.
- Duplicate checks were performed using Order ID + Product ID, with no repeated combinations identified.
- Duplicate rows were removed using Excel's built-in duplicate-removal function.
- The Order Date and Shipping Date fields were removed because the available date values were corrupted/unusable.
- The Ji_lu_shu field was removed because it had no meaningful variation or analytical value for this project.

These checks provided a cleaner and more reliable dataset for the subsequent sales and profitability analysis.

Analysis Methodology

The analysis was conducted in Microsoft Excel using PivotTables, calculated fields, and supporting formulas to examine sales and profitability from multiple perspectives.

The analysis included:

- Category-level analysis to compare total sales, total profit, and profit margin across Furniture, Office Supplies, and Technology.
- Sub-category analysis to identify the products and sub-categories generating the highest and lowest profit margins.
- Discount-band analysis by grouping discounts into four bands: No Discount, Low (1–10%), Medium (11–30%), and High (31%+).
- Regional analysis to compare sales, profit, and profit margins across regions.
- Region × Discount-band analysis to determine how sales and profitability varied by discount level within each region.
- Southeast Asia Tables analysis to isolate the underperforming category and investigate its profitability in greater detail.
- Product-level analysis to identify individual Tables products generating significant losses.
- High-discount sales percentage was calculated to measure the proportion of sales occurring at discounts of 31% or more.
- Profit impact analysis was used to compare profitability at high-discount and no-discount levels.

These analyses were consolidated into an Excel profitability dashboard designed to highlight the main drivers of the category's poor financial performance.

Key Findings

1. Technology generated the strongest overall performance.

Technology recorded the highest sales at $4.74 million and the highest profit at approximately $663,779, making it the strongest-performing product category.

2. Discounting has a significant relationship with profitability.

Across the dataset, the High Discount (31%+) band generated approximately -$793,526 in profit, while transactions with no discount generated approximately $1.77 million in profit.

3. Southeast Asia showed a particularly weak profitability profile.

Southeast Asia generated approximately $884,438 in sales but only $17,852 in profit, resulting in a profit margin of approximately 2.02%.

4. The Tables sub-category in Southeast Asia is loss-making.

Tables generated $52,452 in sales but a loss of $18,618.31, resulting in a -35.50% profit margin.

5. High-discount sales dominate the Southeast Asia Tables category.

Approximately 89.52% of Tables sales were associated with discounts of 31% or more. These high-discount sales generated approximately -$20,144.50 in profit.

6. Losses are concentrated in specific Tables products.

Several individual products recorded substantial losses, including Barricks Computer Table (-$1,965.55), Lesro Training Table (-$1,726.13), and Lesro Training Table, Adjustable Height (-$1,195.24).

7. The problem is therefore not simply low sales.

The analysis indicates that the major concern is the combination of heavy discounting and poor product-level profitability, which is eroding the value generated from Tables sales in Southeast Asia.

Business Recommendations

1. Review and restrict discounts above 31%.

High-discount transactions account for 89.52% of Tables sales in Southeast Asia while generating -$20,144.50 in profit. Management should review the commercial justification for discounts above 31% and introduce approval thresholds for deep discounts.

2. Reassess pricing for loss-making Tables products.

Products with consistently negative profitability should be reviewed for pricing, procurement cost, and discount structure. Where appropriate, prices should be adjusted to restore positive margins.

3. Prioritize margin over sales volume when evaluating promotions.

The analysis shows that generating sales does not necessarily translate into profitability. Promotional decisions should therefore consider profit margin and profit contribution, rather than sales revenue alone.

4. Investigate the economics of the highest-loss products.

Management should investigate the cost structure and pricing of products such as Barricks Computer Table and Lesro Training Table, which recorded some of the largest losses.

5. Establish ongoing discount and profitability monitoring.

A recurring dashboard should track sales, profit margin, discount levels, and loss-making products so that management can identify deteriorating profitability early.

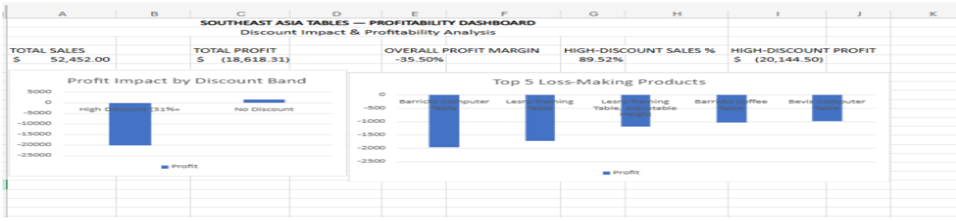
Implementing these recommendations could help the business reduce margin erosion, improve pricing discipline, and redirect promotional activity toward products and discount levels that generate sustainable profit.

The analysis also provides management with a framework for monitoring the relationship between discounting, sales performance, and profitability in future periods.

Dashboard

An interactive Excel dashboard was developed to summarize the profitability analysis and highlight the key drivers of the performance issue.

The dashboard focuses on the Southeast Asia Tables category and presents key performance indicators, discount-level profitability, and the top loss-making products. It enables decision-makers to quickly identify the relationship between discounting and profitability and focus attention on the products requiring further review.



Tools & Skills Demonstrated

Microsoft Excel: Data profiling, data cleaning, PivotTables, calculated fields/formulas, profitability analysis, KPI development, data visualization, and dashboard design.

Analytical Skills: Data validation, segmentation, comparative analysis, profitability analysis, discount-impact analysis, product-level analysis, business insight generation, and recommendations.

Business Skills: Revenue and profit interpretation, margin analysis, identifying profitability drivers, and translating data findings into business recommendations.

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